# AI-Powered Document Analysis & REST API Service

## Executive Summary

This report presents a production-oriented AI-powered document analysis service that ingests structured business documents (CSV/Excel), performs data cleaning using pandas, applies Large Language Model (LLM)-based analysis, validates outputs, and exposes structured insights through a FastAPI REST endpoint. The solution demonstrates modular architecture, schema validation, logging, prompt engineering discipline, and enterprise-ready design principles.

## Problem Statement

Organizations generate vast volumes of structured and semi-structured documents such as invoices, purchase orders, and transaction logs. Extracting insights from such documents traditionally requires manual analytics workflows. The objective of this system is to: 1. Ingest CSV/Excel files 2. Clean and normalize data 3. Generate insights using an LLM 4. Validate output 5. Expose results via REST API

## Dataset Selection

Dataset Used: Online Retail II (Kaggle) https://www.kaggle.com/datasets/mashlyn/online-retail-ii-uci This dataset simulates real-world invoice exports, includes noise and nulls, and supports realistic revenue and operational analytics.

## System Architecture

Client Upload → FastAPI Endpoint → Data Cleaning (pandas) → Aggregation → LLM Analysis → Validation Layer → Structured JSON Response Key architectural principle: Never send raw data to LLM. Perform deterministic calculations first.

## Code Architecture Overview

The project follows modular structure: - config.py: Environment configuration - logger.py: Centralized logging - schemas.py: Pydantic schema definitions - data_cleaner.py: Data normalization - llm_service.py: LLM interaction - qa_validator.py: Output validation - main.py: FastAPI endpoint Separation of concerns ensures maintainability and scalability.

## Data Cleaning Logic Explanation

The cleaning process includes: - Dropping null invoices, quantities, and prices - Converting InvoiceDate to datetime - Removing negative quantities (returns) - Calculating TotalAmount = Quantity × Price This ensures deterministic and accurate revenue computation before sending structured aggregates to the LLM.

## LLM Prompt Engineering Strategy

The LLM is used for reasoning and narrative generation only. Prompt design includes: - Role assignment (senior financial analyst) - Structured input JSON - Strict JSON schema enforcement - Low temperature (0.3) for deterministic output This minimizes hallucination and increases reliability.

## Validation Layer

The validation layer ensures: - Revenue is positive - Recommendations exist - No malformed outputs Schema validation via Pydantic ensures correct structure before API response is returned.

## Production-Readiness Considerations

- Environment-based configuration (.env) - Centralized logging - Schema enforcement - Guardrails against malformed input - Modular architecture - Cost-efficient token usage - Enterprise extensibility

## Conclusion

This AI-powered document analysis system demonstrates a production-oriented integration of data engineering, LLM reasoning, API development, and validation logic. The architecture is scalable, extensible, and suitable for enterprise deployment scenarios.